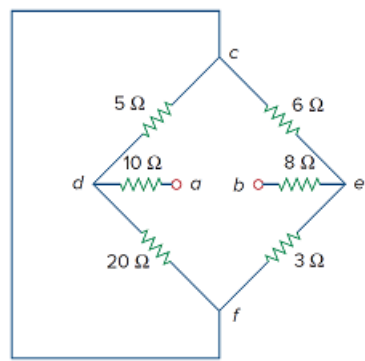


## Problem

Find the equivalent resistance  $R_{ab}$  in the circuit of Fig. 2.111.

Figure 2.111:



## Step-by-step solution

## Step 1 of 7

(a)

Draw the following circuit diagram:

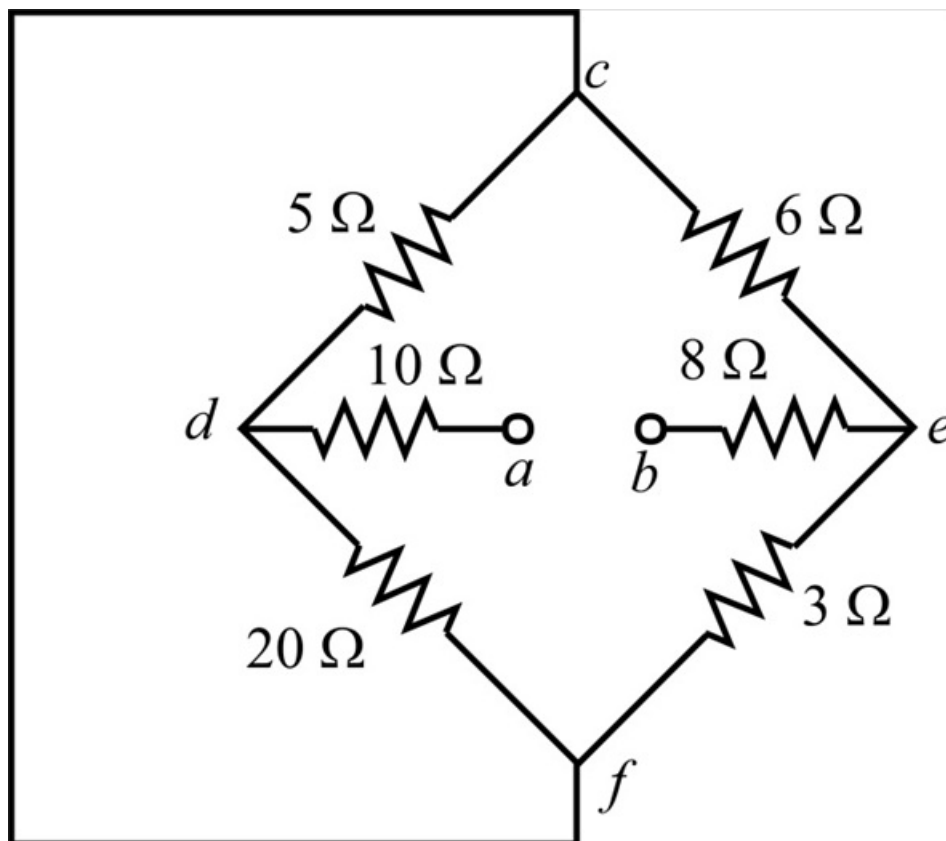


Figure 1

## Step 2 of 7

Clearly from Figure 1, the nodes  $c, f$  are at the same point.

Hence, redraw the Figure 1 as shown in Figure 2.

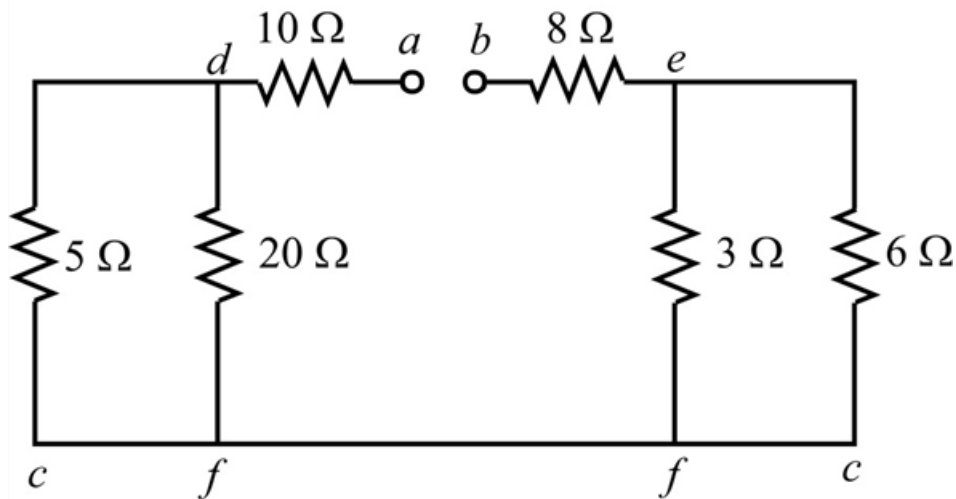


Figure 2

### Step 3 of 7

Clearly from Figure 2,  $20\ \Omega$ ,  $5\ \Omega$  and  $6\ \Omega$ ,  $3\ \Omega$  are in parallel.

Calculate the value of the parallel equivalent resistance  $R_{eq1}$  for  $20\ \Omega$ ,  $5\ \Omega$ .

Write the expression for the parallel equivalent resistance  $R_{eq1}$ .

$$\begin{aligned} R_{eq1} &= 20\ \Omega \parallel 5\ \Omega \\ &= \frac{20\ \Omega(5\ \Omega)}{20\ \Omega + 5\ \Omega} \\ &= \frac{100}{25}\ \Omega \\ &= 4\ \Omega \end{aligned}$$

Therefore, the value of  $R_{eq1}$  is  $4\ \Omega$ .

### Step 4 of 7

Calculate the value of the parallel equivalent resistance  $R_{eq2}$  for  $6\ \Omega$ ,  $3\ \Omega$ .

Write the expression for the parallel equivalent resistance  $R_{eq2}$ .

$$\begin{aligned} R_{eq2} &= 6\ \Omega \parallel 3\ \Omega \\ &= \frac{6\ \Omega(3\ \Omega)}{6\ \Omega + 3\ \Omega} \\ &= \frac{18}{9}\ \Omega \\ &= 2\ \Omega \end{aligned}$$

Therefore, the value of  $R_{eq2}$  is  $2\ \Omega$ .

### Step 5 of 7

Redraw the modified Figure 2 as shown in Figure 3.

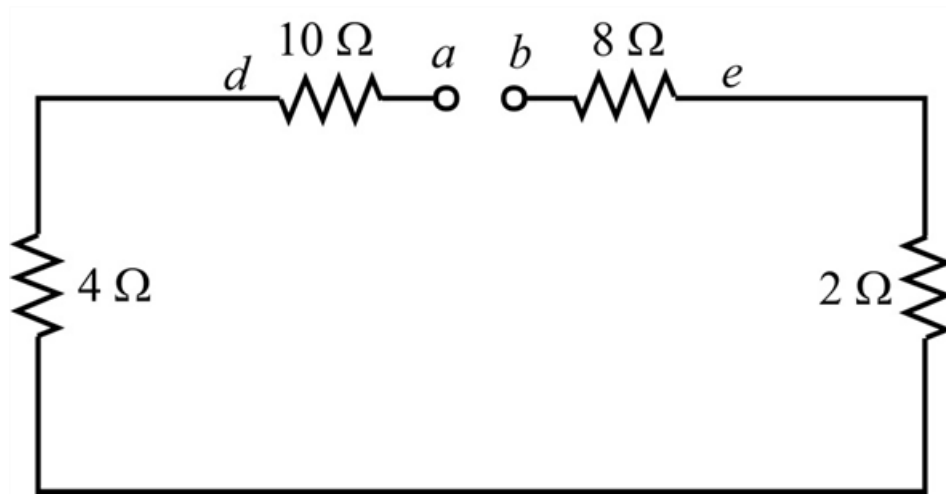


Figure 3

**Step 6** of 7

Clearly from Figure 3,  $10\ \Omega, 8\ \Omega, 2\ \Omega, 4\ \Omega$  are in series respectively.

Calculate the value of the series equivalent resistance  $R_{ab}$  for  $10\ \Omega, 8\ \Omega, 2\ \Omega, 4\ \Omega$ .

Write the expression for the series equivalent resistance  $R_{ab}$  for  $10\ \Omega, 8\ \Omega, 2\ \Omega, 4\ \Omega$ .

$$\begin{aligned} R_{ab} &= 10\ \Omega + 8\ \Omega + 2\ \Omega + 4\ \Omega \\ &= 24\ \Omega \end{aligned}$$

Therefore, the value of the series equivalent resistance  $R_{ab}$  is  $24\ \Omega$ .

Redraw the modified Figure 3 as shown in Figure 2.

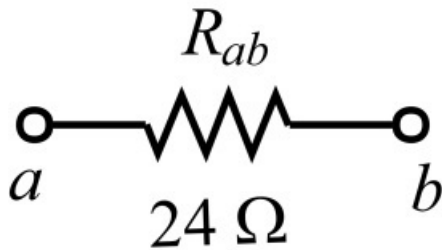


Figure 3

**Step 7** of 7

Clearly from Figure 3, the value of required equivalent resistance  $R_{ab}$  is  $24\ \Omega$ .